Kunal Singh

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PROFILE

Cyber Security undergraduate who builds security tools from first principles in C, C++, Java and Python, applying data structures, algorithm design, operating systems and database internals to real defensive problems: tamper-evident logging, syscall sandboxing, injection detection and attack-path analysis.

EDUCATION

B.Tech in Cyber Security · CGPA: 8.2 / 10 — 07/2024 – 07/2028 | Chennai
Vellore Institute of Technology, Chennai

Senior Secondary (Class XII) — 05/2024 | Ludhiana
D.A.V Public School, Ludhiana

Secondary (Class X) — 05/2022 | Ludhiana
D.A.V Public School, Ludhiana

TECHNICAL SKILLS

Languages	C, C++, Java, Python, SQL
Core CS	Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, DBMS
Security	Honeypots (Cowrie), syscall tracing & sandboxing, SQL-injection detection, hash-based integrity (SHA-256, Merkle trees)
Tools	Linux, Git, SQLite / MySQL

PROJECTS

ChainLog – Tamper-Evident Audit Log Engine — **C++ · SHA-256 · Merkle Trees**

- Built an append-only log store in C++ where every record is hash-chained and batched into a Merkle tree, so any edit, deletion or reordering of history is detectable.
- Implemented O(log n) inclusion proofs, a write-ahead append path for crash safety (borrowed from DBMS recovery) and a verifier CLI that pinpoints the first corrupted record.

SandboxLite – Syscall-Level Process Sandbox — **C · Linux ptrace / seccomp · Python**

- Built a Linux sandbox that runs untrusted binaries under ptrace and seccomp, logs every system call and blocks anything outside a per-program allowlist (e.g. *execve*, *socket*, *ptrace*).
- Wrote self-made test programs that attempt file, network and process escapes, and a Python report that flags runs whose syscall sequences deviate from a learned baseline.

QueryShield – SQL-Injection Detection Proxy — **Java · SQL · Aho–Corasick**

- Built a Java reverse proxy that screens requests with a SQL tokenizer and an Aho–Corasick multi-pattern matcher, scanning payloads in linear time.
- Tested it in front of a deliberately vulnerable MySQL-backed app I wrote, using a labelled attack/benign query set, and tuned rules to cut false positives on legitimate queries.

ChokePoint – Attack-Path & Patch-Priority Analyzer — **Python · Dijkstra · Max-Flow / Min-Cut**

- Models a network as a weighted directed graph (hosts, services, vulnerability-based edge costs) and uses Dijkstra to find the cheapest attacker route from an entry point to a critical asset.
- Uses max-flow / min-cut to find the smallest set of hosts or links whose fix severs every attack path, and outputs a ranked patch list.

ShadowPot – SSH Honeypot & Attacker Analytics — **Python · Cowrie · SQL**

- Deployed a Cowrie-based SSH honeypot and loaded its session logs into a normalized SQL schema; wrote analytics queries on credentials tried, source IPs and post-login commands to study attacker behaviour.

Disaster Resource Allocation System — **SQL · DBMS**

- Designed a normalized relational schema for relief resources and used SQL triggers to automate allocation and keep inventory consistent.

COURSES

Introduction to Cybersecurity — *Cisco Networking Academy* — 01/2026 – 02/2026
Database Management Essentials — *Coursera (University of Colorado Boulder)* — 02/2026 – 03/2026
AI For Everyone — *Coursera (DeepLearning.AI)* — 01/2026 – 02/2026